

Advanced Signal Processing

Lecture notes

Signal Spaces

- Vector spaces
- Inner product spaces
- Signal spaces
- Projection of a signal on signal
- Bases
- Orthonormal bases
and vector isomorphism.
- Exercises.

Vector Space

A vector space V is a collection of elements $v \in V$ called vectors, along with 8 axioms:

1) Commutativity: $v + u = u + v$

2) Associative:

$$v + (u + w) = (v + u) + w$$

3) Zero: Unique $0 \in V$ st. $0 + v = v$
for all $v \in V$.

4) Negation: for each $v \in V$ there is
a unique $-v \in V$ st. $v + (-v) = 0$.

5) Multiplication with scalar is associative
 $\alpha(\beta v) = (\alpha\beta)v$, α, β scalars
 $v \in V$

6) Unit multiplication: $1v = v$

7) $\alpha(v + u) = \alpha v + \alpha u$.

8) $(\alpha + \beta)v = \alpha v + \beta v$.

Examples of vector spaces

- The set of n -tuples $(\mathbb{C}^n)$
 $\underline{x} = (x_1, x_2, \dots, x_n) \in U$
- The set of infinite sequences
 $\underline{x} = (x_1, x_2, x_3, \dots) \in U$
- Real-valued functions
 $U = \{ \underline{x} : \text{function } \mathbb{R} \rightarrow \mathbb{R} \}$
- Complex-valued functions
- Random variables
- Stochastic processes

And many more!

Inner product space

An inner product space is a vector space V equipped with an inner product $\langle \cdot, \cdot \rangle$

1) $\langle \underline{x}, \underline{y} \rangle = \langle \underline{y}, \underline{x} \rangle^*$ [Symmetry]

2) $\langle \underline{v} + \underline{w}, \underline{y} \rangle = \langle \underline{v}, \underline{y} \rangle + \langle \underline{w}, \underline{y} \rangle$
and $\langle \alpha \underline{v}, \underline{y} \rangle = \alpha \langle \underline{v}, \underline{y} \rangle$

[Linearity]

3) $\langle \underline{x}, \underline{x} \rangle \geq 0$, equal iff. $\underline{x} = 0$
[positivity]

Norm of a vector

The norm of a vector $x \in U$:

$$\|x\| = \sqrt{\langle x, x \rangle}$$

Orthogonal Vectors

Vectors $x, y \in U$ are orthogonal if

$$\langle x, y \rangle = 0$$

Signal space

A signal space is an inner-product space where all elements have finite norm. Each element is called a signal.

Remark:

In signal processing $\|x\|^2$ is called the energy of x .

Thus elements of a signal space have "finite energy" by definition. We also speak about a "finite energy signal", which is strictly speaking a tautology.

Examples of signal spaces:

- Finite-length real-valued discrete-time signals $\underline{x} = [x_1, x_2, \dots, x_N]^T$, $\underline{y} = [y_1, \dots, y_N]^T$

$$\langle \underline{x}, \underline{y} \rangle = \sum_{n=1}^N x_n y_n = \underline{x}^T \underline{y} = \underline{y}^T \underline{x}$$

$$\|\underline{x}\|^2 = \langle \underline{x}, \underline{x} \rangle = \sum_n x_n^2$$

- Complex ditto:

$$\langle \underline{x}, \underline{y} \rangle = \sum_{n=1}^N x_n y_n^*$$

$$\|\underline{x}\|^2 = \sum_n |x_n|^2$$

- Real valued square integrable functions

$x, y \in U$ are $\mathbb{R} \rightarrow \mathbb{R}$ functions

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y(t) dt$$

$$\|x\|^2 = \langle x, x \rangle = \int_{-\infty}^{\infty} x^2(t) dt$$

- Complex ditto

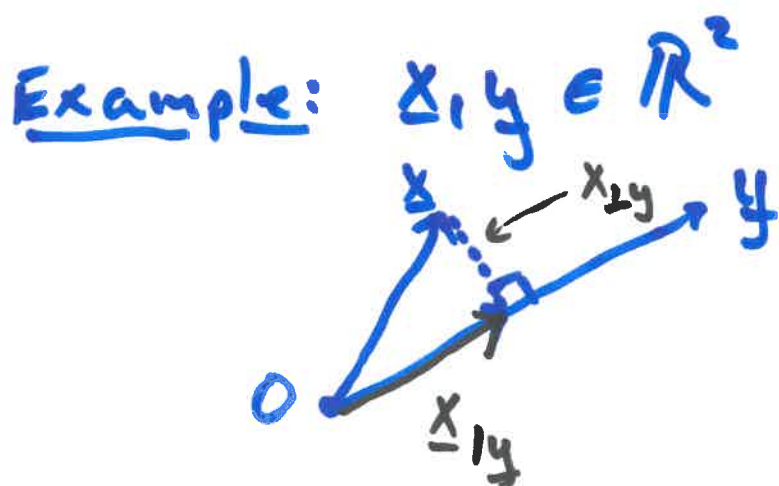
Projection of signal on signal

Projection of signals is defined in the same manner as in Linear Algebra:

Projection of signal x onto signal y :

$$\underline{x}_{1y} = \frac{\langle x, y \rangle}{\|y\|^2} y$$

Note: If $\|y\|^2 = 1$, then $\underline{x}_{1y} = \langle x, y \rangle y$



$$\underline{x}_{\perp y} = x - \underline{x}_{1y}.$$

Base for Signal Space

A collection of signals $\Phi = \{\phi_1, \phi_2, \phi_3, \dots\}$ is called a base for signal space U if Φ spans U :

$$U = \text{span}\{\Phi\}$$

$$= \{ \text{all linear combinations of } \phi_1, \phi_2, \dots \}$$

$$= \left\{ \sum_n a_n \phi_n : \text{for all scalars } a_1, a_2, \dots \right\}$$

Remark

If U is $\mathbb{R}^N$ or $\mathbb{C}^N$, each ϕ_n is often called a basis vector.

If instead U is the space of real or complex functions, ϕ_n are called basis functions.

Orthonormal Basis

An orthonormal Basis fulfills:

$$\Phi = \{\phi_1, \phi_2, \dots\} \text{ where}$$

$$\langle \phi_i, \phi_j \rangle = \begin{cases} 1 & i=j \text{ (normalized)} \\ 0 & i \neq j \text{ (orthogonal)} \end{cases}$$

To represent ^{signal} $\underline{x}$ in the basis:

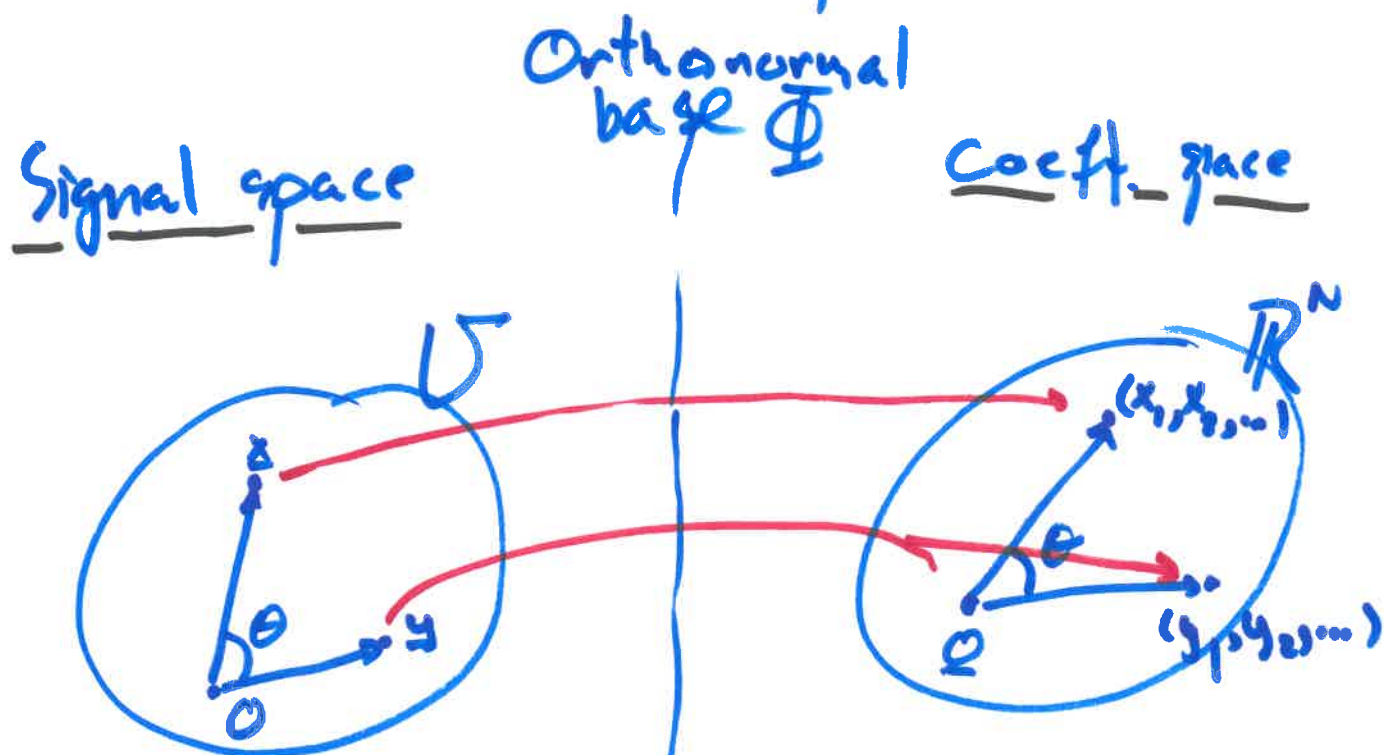
$$x_n = \langle \underline{x}, \phi_n \rangle \quad n=1, 2, \dots$$

To recover $\underline{x}$ from $x_1, x_2, \dots$

$$\underline{x} = \sum_n x_n \phi_n \quad (\text{synthesis eq.})$$

Vector Isomorphism

The vector space U and the space of coefficients ($\mathbb{R}^N$ or $\mathbb{C}^N$) are connected - in fact they are isomorphic. This means that norms and inner products are preserved.



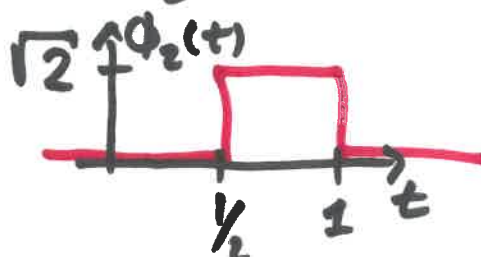
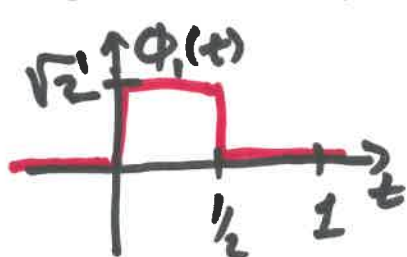
$$\langle x, y \rangle = \sum_n x_n y_n \quad (\text{Inner prod preserved})$$

$$\|x\|^2 = \langle x, x \rangle = \sum_n |x_n|^2 \quad (\text{norm preserved})$$

Exercise

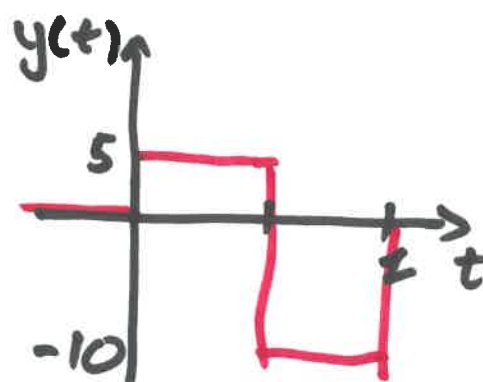
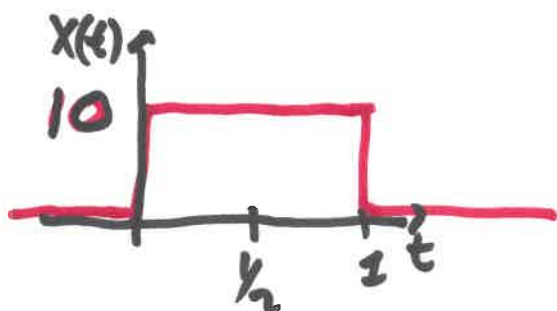
Consider a signal space $\mathcal{U}$ with basis

$$\Phi = \{ \phi_1(t), \phi_2(t) \}$$



1) Confirm that Φ is an orthonormal set, i.e. check that ϕ_1 and ϕ_2 are orthogonal and normalized.

2) Check that the two signals $x(t), y(t)$ are in $\mathcal{U}$:



3) Give an example of a signal defined on the time interval $t \in [0, 1]$ which is not in $\mathcal{U}$.

4) Represent $x(t)$ and $y(t)$ in the basis $\mathcal{F}$. (compute x_1, x_2 and y_1, y_2)

5) Evaluate $\langle \underline{x}, \underline{y} \rangle$ and compare this to $\sum_n x_n y_n$.

Similarly compare $\|\underline{x}\|^2$ to $\sum_n x_n^2$.

6) Construct a signal $z(t)$ in V which is orthogonal to $x(t)$, i.e.

$$\langle \underline{z}, \underline{x} \rangle = 0$$

Check that $\sum_n z_n x_n = 0$